

Lab Program 8

Title: Create Kubernetes Pods using YAML Descriptors

Project Structure:

prog8/

nodejs-configmap.yaml

nodejs-pod.yaml

[server.js](#)

STEP 1: Start Minikube

\$ minikube start

```
victus% minikube start
🐸 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👍 Starting "minikube" primary control-plane node in "minikube" cluster
🚧 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
```

STEP 2: Check the status of Minikube

\$ minikube status

```
victus% minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

STEP 3: Write Node.js app

\$ nano [server.js](#)

```
const http = require("http");
const port = 3000;

const server = http.createServer( (req, res) => {
    res.end("Nodejs running inside Kubernetes!");
});

server.listen(port, () => {
    console.log(`server running at port ${port}`);
});
```

STEP 4: Write a config map file

\$ nano nodejs-configuration.yaml

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: nodejs-app-config
data:
  server.js: |
    const http = require("http");
    const port = 3000;

    const server = http.createServer((req, res) => {
      res.end("Nodejs running inside Kubernetes!");
    });

    server.listen(port, () => {
      console.log(`server running at port ${port}`);
    });
```

STEP 5: Write a Pod YAML file using [Node.js](#) as a base image

```
$ nano nodejs-pod.yaml
```

```
apiVersion: v1
kind: Pod
metadata:
  name: nodejs-app-pod
  labels:
    app: nodejs
spec:
  containers:
  - name: nodejs
    image: node:18-alpine
    command: ["node", "/usr/src/app/server.js"]
    volumeMounts:
    - name: app-code
      mountPath: /usr/src/app
    ports:
    - containerPort: 3000
  volumes:
  - name: app-code
    configMap:
      name: nodejs-app-config
```

STEP 6: Apply the YAML files

```
$ kubectl apply -f nodejs-configuration.yaml
```

```
victus% kubectl apply -f nodejs-configmap.yaml
configmap/nodejs-app-config configured
```

```
$ kubectl apply -f nodejs-pod.yaml
```

```
victus% kubectl apply -f nodejs-pod.yaml
pod/nodejs-app-pod configured
```

STEP 7: Check the status of the Pod

\$ kubectl get pods

```
victus% kubectl get pods
```

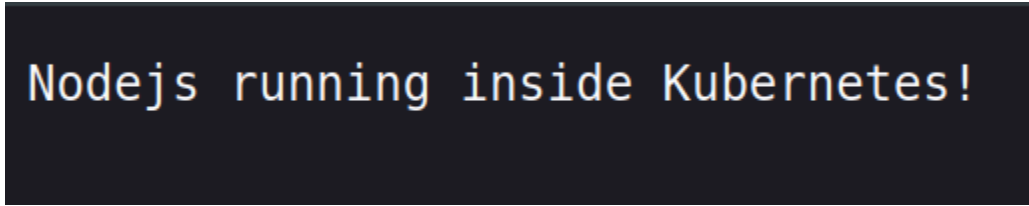
NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f9664446b-pprxm	1/1	Running	0	26m
nginx-deployment-6f9664446b-rj4pl	1/1	Running	0	26m
nodejs-app-pod	1/1	Running	0	9m51s

STEP 8: Forward the port to access the node application

\$ kubectl port-forward pod/nodejs-app-pod 3000:3000

```
victus% kubectl port-forward pod/nodejs-app-pod 3000:3000
Forwarding from 127.0.0.1:3000 → 3000
Forwarding from [::1]:3000 → 3000
Handling connection for 3000
```

STEP 9: Access the application in the URL: <http://localhost:3000>



Nodejs running inside Kubernetes!

STEP 10: Delete Pod

\$ kubectl delete pod nodejs-app-pod

```
victus% kubectl delete pod nodejs-app-pod
pod "nodejs-app-pod" deleted
```